

Agent evaluation results: long-horizon software tasks

Sheil Sarda

September 2026

Executive summary

- Evaluated 11 tasks across 3 models and 3 effort levels: 99 first counted results, with 76 passes.
 - Of those successes, 64 exceeded the take-home's 75-step threshold, measured as assistant turns.
 - Counted failures include service interruptions.
 - Selected three Harbor tasks with observed headroom in Sonnet:
 - Rerun: chunk optimizer. Sonnet passed its own checks but returned chunks containing 6 and 2 rows instead of the required 4 and 4.
 - Zenoh: timestamp instrumentation. Sonnet reported successful checks but exposed the wrong public identifier type, causing downstream compilation failures.
 - Burn: PyTorch checkpoint reader. Sonnet missed parsing and bounds requirements. A separate attempt passed functional checks but violated a source restriction.
 - Across these three tasks, Fable and Opus passed 18/18 combined. Sonnet passed 5/9, showing failures on requirements that other agents successfully satisfied.
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Executive summary: task selection

Rerun, Zenoh timestamps and Burn checkpoint reader

Candidate	Pass / first trials	Median turns on success	Reason to retain
Rerun chunk optimizer	8/9	119	Chunk layout and data preservation
Zenoh timestamps v3	8/9	216.5	Wire protocol and public API integration
Burn checkpoint reader v4	7/9	144	Format compatibility and bounds checks

Sonnet failed four of the nine sample settings

Fable and Opus: 18/18 passes. Sonnet: 5/9 passes.

We retain separate records for repeats and corrected Zenoh regrades.

Software integration across modules

11 completed task cohorts across C++, Rust and Python

Implementing features across APIs, data formats and storage systems

99/99

combinations with counted
results

64/76

successful trials above 75
assistant turns

11

evaluated tasks, with three
detailed cases

Feature requirements across three languages

C++

Extend callback, playback and connectivity APIs while preserving compatibility and observable behavior.

Python

Implement storage writers, codecs and persistence features with explicit recovery and representation contracts.

Rust

Integrate checkpoint readers, runtime weights, streaming layouts and timestamp metadata across modules.

Agents must preserve the specified behavior across module boundaries.

Passes in the first result grid

Passes / counted results.

Task	Fable	Opus	Sonnet	Covered
Foxglove parameter handler	3/3	3/3	3/3	9/9
rosbag2 mixed playback	3/3	2/3	2/3	9/9
Zenoh C++ connectivity	3/3	2/3	3/3	9/9
Diskcache resharding v2	1/3	1/3	0/3	9/9
Luigi generation target	1/3	3/3	0/3	9/9
Rosbags storage writers	3/3	3/3	3/3	9/9
Zarr value codecs	2/3	3/3	1/3	9/9
Burn ONNX runtime weights	3/3	3/3	2/3	9/9
Burn checkpoint reader v4	3/3	3/3	1/3	9/9
Rerun chunk optimizer	3/3	3/3	2/3	9/9
Zenoh timestamps v3	3/3	3/3	2/3	9/9

Service interruptions affected Luigi and Diskcache.

Results by model and effort (1 of 2)

One counted result per task, model and effort.

Task	Fable Med	Fable High	Fable Max	Opus Med	Opus High	Opus Max	Sonnet Med	Sonnet High	Sonnet Max
Foxglove parameters	P (79)	P (134)	P (191)	P (82)	P (133)	P (168)	P (163)	P (270)	P (364)
rosvbag2 mixed playback	P (112)	P (146)	P (194)	F (99)	P (125)	P (155)	P (154)	F (251)	P (381)
Zenoh C++ connectivity	P (97)	P (115)	P (184)	P (62)	P (85)	F (134)	P (87)	P (165)	P (331)
Diskcache resharding v2	P (70)	F (74)	F (24)*	F (59)	P (117)	F (143)	F (60)	V (83)	A (39)
Luigi generation target	F (67)	P (41)	A (20)*	P (47)	P (57)	P (110)	F (63)	F (96)	A (32)
Rosbags storage writers	P (20)	P (44)	P (90)	P (22)	P (29)	P (80)	P (43)	P (53)	P (173)

P pass / F scored failure / A agent timeout / V verifier timeout. (n) turns. * Service affected. Repeated trials have separate records.

Results by model and effort (2 of 2)

One counted result per task, model and effort.

Task	Fable Med	Fable High	Fable Max	Opus Med	Opus High	Opus Max	Sonnet Med	Sonnet High	Sonnet Max
Zarr value codecs	F (63)	P (93)	P (153)	P (88)	P (105)	P (96)	F (117)	F (142)	P (335)
Burn ONNX weights	P (162)	P (194)	P (260)	P (134)	P (230)	P (220)	F (217)	P (288)	P (446)
Burn checkpoint reader v4	P (92)	P (138)	P (158)	P (76)	P (144)	P (158)	F (42)	F (251)	P (323)
Rerun chunk optimizer	P (115)	P (110)	P (154)	P (72)	P (104)	P (123)	F (138)	P (193)	P (383)
Zenoh timestamps v3	P (166)	P (188)	P (216)	P (176)	P (217)	P (245)	F (377)	P (474)	P (626)

P pass / F scored failure / A agent timeout / V verifier timeout. (n) turns. * Service affected. Repeated trials have separate records.

C++: API and playback integration

One attempt per model x effort combination (3 x 3 = 9 attempts per task)

Task / first results	Required behavior	Interpretation
Foxglove parameters 9/9 pass, 163 turns	Expose delayed responses through Rust, C FFI and C++ while preserving ownership.	All nine pass above 75 turns. Sustained work, with no observed failure in this first cohort.
rosbag2 mixed playback 7/9 pass, 154 turns	Play mixed serialization formats and filter topics while preserving raw rewrite behavior.	One failure violates warning rules. The other fails an ambiguous writer assertion. Treat them separately.
Zenoh C++ connectivity 8/9 pass, 106 turns	Wrap transport and link events across two C++ backends with ownership and callback parity.	The only failure is an upstream segfault. New connectivity checks pass. Its cause remains unresolved.

Turns are the median among successful trials.

Python: persistence and format contracts

Task / first results	Required behavior	Interpretation
Diskcache resharding 2/9 pass, 93.5 turns	Reshard live caches with crash recovery and mutation reconciliation.	Timeouts and quota affect the grid. Some failures resurrect deleted data.
Luigi generation target 4/9 pass, 52 turns	Publish atomic file generations with stable snapshots and crash recovery.	Some lifecycle rules fail. One of the two timeouts is gateway affected.
Rosbags storage writers 9/9 pass, 44 turns	Write SQLite3 and MCAP bags with metadata, compression and interoperability.	All nine pass. Only 3/9 successes exceed 75 turns. No failures in these nine attempts.
Zarr value codecs 6/9 pass, 100.5 turns	Compose cast and scale/offset codecs with exact metadata and integer semantics.	Cast-value parity passes. Validation, JSON and integer scale/offset failures remain.

Turns are the median among successful trials.

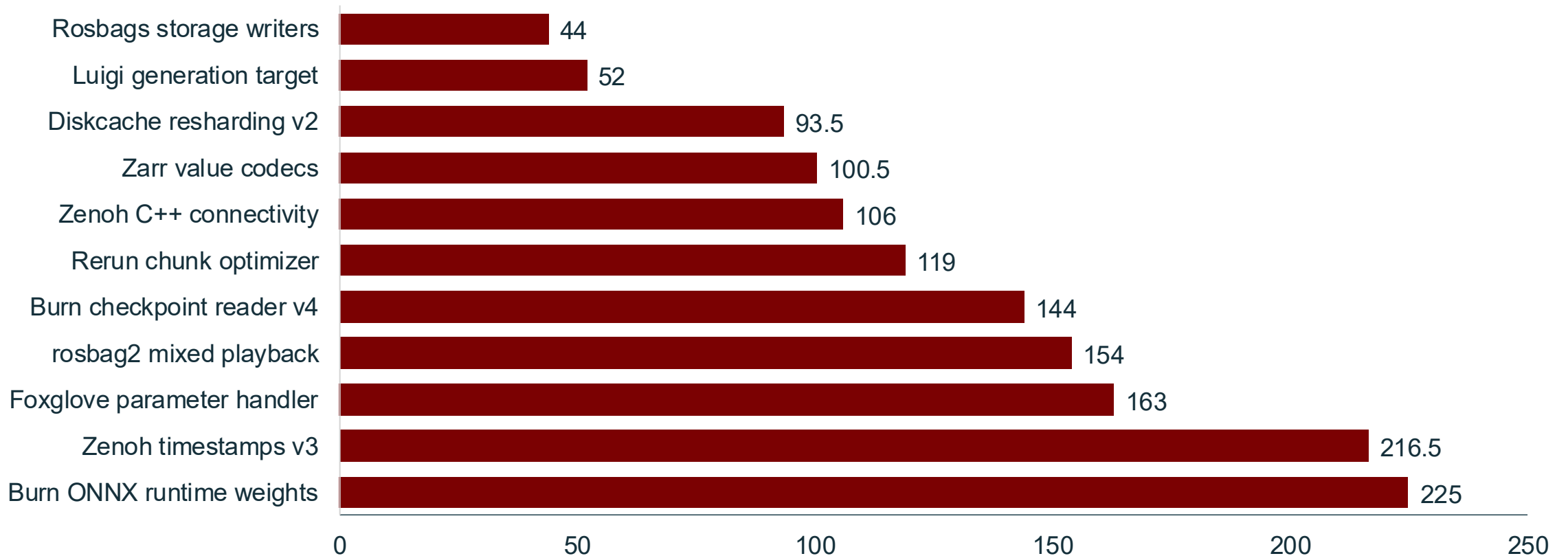
Rust: runtime behavior and regression failures

Task / first results	Required behavior	Interpretation
Burn ONNX weights 8/9 pass, 225 turns	Use runtime RNN weights while preserving layouts, outputs and initializers.	Eight pass above 75 turns. The failure is a snapshot regression. Runtime checks pass.
Burn checkpoint reader 7/9 pass, 144 turns	Read checkpoint data on demand, reject unsafe input and honor source constraints.	One functional failure and one source restriction failure. Sonnet max passes.
Rerun chunk optimizer 8/9 pass, 119 turns	Stream and repack chunks while preserving cells, memory bounds and layout.	One required layout failure. Sonnet max passes after a longer trajectory.
Zenoh timestamps v3 8/9 pass, 216.5 turns	Carry timestamps through public APIs, wire encoding and receive paths.	The same public-type failure persists. All eight other saved regrades pass.

Turns are the median among successful trials.

84% of successful runs exceeded take-home's 75-step threshold

Median turns on success. 64/76 successful trials exceed 75 turns



Rerun: incorrect chunk layout

Returned layout, rows per chunk



Required layout, rows per chunk



“All existing tests for the three crates pass, plus my own smoke tests”

Sonnet failed five split, packing and row-limit checks while passing the existing regression suites.

Sonnet attempts on the same task	Observed result
sonnet/medium, 138 steps, fail	Layout contract failures
sonnet/max, 383 steps, pass	All verifier groups passed

Zenoh: Sonnet used the wrong public type

Corrected verifier: eight passes and the same API failure in nine saved submissions

Required public type

The instruction requires ZenohId for TimestampContext.zid. Sonnet used the wire type ZenohIdProto.

Downstream compilation

Downstream users compile against the specified API. The type mismatch prevents that code from compiling.

“Everything is green with both feature configurations.”

Sonnet effort	Observed result
Medium, 377 steps	Type mismatch fails both verifiers
Max, 626 steps	Passes both v3 and corrected verifiers

Burn: parser failures and a source restriction

Observation	Interpretation
Sonnet medium, 42 steps	Parser and validation failures
Sonnet high, 251 steps	Functional pass, source restriction failure
Sonnet max, 323 steps	Passes all verifier groups
Fable / Opus max, 158 steps each	Both pass all verifier groups

Seven successful attempts took 76–323 turns. Failure causes include parser behavior and a source restriction.

Failure review and test corrections

The held SQLite task is outside the 11-task result grid

NUL defaults in SQLite

The task's all-strings requirement includes NUL defaults. All 30 attempts failed that test. Three failed no other checks.

Corrections to hidden tests

Reviewers corrected hidden internal imports, an unstated Debug bound and an empty-rename rejection. We record revised tasks and saved-submission regrades separately.

Keep separate counts for implementation defects, source restrictions, test/spec mismatches and infrastructure failures.

Our Rerun and Zenoh cases show agents declaring success while required behavior remained incorrect.

20

tasks in SWE-Marathon

<30%

solved by agents evaluated in the paper

“poor self-verification, self-reported infeasibility, and premature termination”¹

We compare submissions with the feature requirements and the results of local checks.

1. Desai et al., “SWE-Marathon: Can Agents Autonomously Complete Ultra-Long-Horizon Software Work?” (abstract), arXiv:2606.07682, 2026.

Pilot findings and measurement limits

Observed task completions

Agents passed 76 of 99 first counted results. In 64 successful attempts, they used more than 75 turns. Reviewers traced several failures to stated requirements.

Measurements for another study

We need repeated trials to estimate pass rates. We cannot rank models or attribute changes to effort. We have not measured human time or training gains.

Service interruptions affect the Fable/max outcomes in Luigi and Diskcache.

SWE-Bench Pro: “patches across multiple files and substantial code modifications.”

METR uses measured human completion time. We count assistant turns without a human baseline.

Scaling to 1,000 accepted tasks

01

**Mine issues and
PR stacks**

02

**Compose
dependent
changes**

03

**Audit
requirements and
controls**

04

**Hold out task
families**

Public sources: Rerun, Burn, Zenoh, Foxglove, Luigi and Zarr

Pilot batches: 25, then 100, then 1,000 accepted tasks

Track acceptance yield, authoring time, task validity, runtime cost and distinct failure mechanisms at each stage.

Evaluating 1,000 comparable tasks would cost about \$198,000

Cost projections through 1,000 tasks

\$2,914

recorded study spend, 207
counted trials including repeats

\$594

for the 27 first trials of the
three recommended tasks,
\$198 per task

\$198K

projected for 1,000 tasks on
the full grid; \$73,000 for
Sonnet only

These projections assume similar task complexity and API pricing. They exclude task authoring, compute, human review and additional retries.

Production budgeting must include discarded candidates. Measure acceptance yield and total cost per accepted task before committing to a price.

Seven trials have incomplete cost records.

Task pack and next experiment

Evidence for the report

We retain results for all 11 tasks, raw artifact hashes and turn counts. The three sample cases include failed attempts and successful contrasts.

Next experiment

After provider access returns, repeat fixed task/model/effort settings. Estimate pass rates with uncertainty and review recurring failures against the requirements.

We have 99 counted results. Provider usage limits interrupted the sweep.